MODULE 1 · INTRODUCTION

# Lesson 03: Causes and Contributing Factors of Upsets

Lesson 3 of 24 · about 8 min

**LEARNING OBJECTIVES**

- Describe the three main families of upset causes: environmental, systems-induced and pilot-induced.
- Recognise how several factors usually combine in an upset.
- Relate each family of causes to prevention strategies.



Environmental, systems and pilot factors usually combine to cause an upset.

## Three families of causes

| Family | Examples |
|---|---|
| **Environmentally induced** | Turbulence and clear-air turbulence; mountain waves; windshear and microbursts; wake turbulence; thunderstorms; airframe icing; temperature changes at high altitude |
| **Systems-induced** | Autopilot or autothrust behaviour and disconnects; flight control malfunctions or degraded control laws; erroneous air data (unreliable airspeed); instrument failures; asymmetric thrust or engine failure |
| **Pilot-induced** | Inattention and poor monitoring; distraction; inappropriate control inputs; mode confusion; poor energy management; spatial disorientation; startle; fatigue; inadequate manual flying skill |

## Upsets are rarely caused by one factor

Accident investigations show a typical pattern: an **initial trigger** — often environmental or technical — followed by **human responses** that do not contain it, or that make it worse. For example:

- Ice crystals block the pitot probes (environmental) → the autopilot disconnects and air data is unreliable (systems) → the pilot makes sustained nose-up inputs under stress (pilot-induced) → the aircraft stalls.
- A faulty sensor causes the autothrust to reduce thrust (systems) → the crew does not notice the speed decay (monitoring) → the stall warning surprises them at low altitude (startle).

## Prevention strategies for each family

- **Environmental:** knowledge of the threats, weather avoidance, respecting wake separation, briefing and planning, and using the right speeds and altitudes with adequate margins (Lesson 12).
- **Systems:** understanding the automation and its modes, monitoring the flight mode annunciations, knowing the memory items for unreliable airspeed and flight control malfunctions (Lessons 13 and 18).
- **Pilot:** disciplined monitoring, manual handling practice, energy awareness, managing startle, fatigue awareness and CRM (Lessons 14–16).

> **THREAT AND ERROR MANAGEMENT**
> Treat every cause as a **threat** to be anticipated and briefed. Most upset-related accidents could have been stopped at the threat or error stage — long before an upset recovery was needed.

**KEY TAKEAWAYS**

- Causes are environmental, systems-induced or pilot-induced — usually in combination.
- A technical or weather trigger often becomes an upset through human responses.
- Each family of causes has its own prevention strategy.

**KNOWLEDGE CHECK**

**Which is an example of a systems-induced upset cause?**

A. Wake turbulence from a preceding aircraft

B. Unreliable airspeed after a pitot probe blockage

C. A microburst on approach

D. Fatigue during a night flight

**Answer: B.** Erroneous air data from a blocked pitot probe is a systems-induced factor, even if the blockage itself was caused by weather.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.